Annex A
Fluid Test List

1

Table 3A: Fluid Test List - Aircraft and Ground Applications 2

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Test fluid		Specification	Temperature	Notes
Fuels	Aviation turbine	Def Stan 91-91 or Def Stan 91-87	70°C ± 2°C	AVTUR F35 AVTUR FSII F34(JP8)[6]
	Fuel Oils	ISO 1817 Liquid F	70°C ± 2°C	Simulates diesel fuel, domestic heating and similar light furnace oils
	Gasoline	ISO 1817 Liquid B	40°C ± 2°C	
Hydraulic fluids	Mineral based	Def Stan 91-48 (see Note 1)	50°C ± 2°C	OM15
	Phosphate ester based (synthetic)	DTD 900/4881D	100°C ± 5°C	OX 20 Skydrol 500 B4
		ISO 1817 Liquid 103	70°C ± 2°C	
	Silicone Based	Dimethyl silicone, 20 mm^2/s (CST) AT 25°C NATO S1716	70°C ± 2°C	This replaces S1714 which is no longer available
	Non Mineral Based (synthetic)	80% 2 – Ethoxyethanol (Ethylene Glycol Monoethyl Ether) to BS 2713 and 20% Castor Oil to BS 7207 by volume	100°C ± 5°C	
Lubricating oils	Mineral based	SAE J 1899 Grade 50	100°C ± 5°C	
	Ester based (synthetic)	ISO 1817 Liquid 101	150°C ± 5°C	
Solvents and cleaning fluids	Propan-2-ol (isopropyl alcohol)	BS 1595	50°C ± 2°C	
	Detergent cleaning fluid minimum pH 10	MIL-C-43616	20°C ± 2°C	
	Cleaning Fluid	Aliphatic hydrocarbon solvent	20°C ± 2°C	Lotoxane
De-icing and antifreeze fluids	Kilfrost ABC	DTD 900/4907	50°C ± 2°C	
		50% inhibited potassium acetate in water	20°C ± 2°C	
Sullage		4 ± 0.1% formaldehyde plus 1 ± 0.1 % Cresylic acid (mixed isomers general purpose reagent) and 95% water	23°C ± 2°C	(see Note 2)
Salt water		3.5% NaCl	50°C ± 2°C	
De-ionised water			50°C ± 2°C	

NOTE 1 NATO H-515 may be used as an alternative.

NOTE 2 If difficulty is experienced in producing a solution, pure m-cresol may be used

NOTE 3 Some test temperatures exceed the critical flash point temperature. Expert advice shall be sought on the conduct of these tests.

NOTE 4 Tests at elevated temperatures shall be carried out at the temperatures shown or the cable rated temperature which ever is the lowest unless otherwise specified in the product standard for the purpose of short term exposure assessment

Note 5 Exposure periods shall be 24 hours unless otherwise specified in the product standard

Note 6 AVTUR based fuels F 34 and F35 are not acceptable for embarkation on HM Ships

Table 3 B – Fluids Test List - Marine Applications 1

2

Test fluid		Specification	Temperature	Notes
Fuel Oil	Dieso	NATO F-76 DEF STAN 91-4/Issue 5(or DERV)	20 ± 3°C	
Aviation Turbine Fuel		NATO F-44 AVCAT/FSII (JP 5)	70°C ± 2°C	
Hydraulic Fluid	Petroleum based	OX 30 Def Stan 91-35	50 ± 3°C	
		Water/Glycol based OX 40 Def Stan 91-110	50 ± 3°C	
Lubricating Oil	Detergent Mineral	OMD-113	50 ± 3°C	
	Synthetic ester base	OX-28 (DERD 2499)	50 ± 3°C	
Water	Water	Deionized	50 ± 3°C	
	Salt water	Deionized with 3½% Na Cl	50 ± 3°C	
Note 1 For marine cable tests it is normal to conduct long term exposure tests 7 days for cores and 28 days for outer jackets or sheathes				
Note 2 Tests at elevated temperatures shall be carried out at the temperatures shown or the cable rated temperature which ever is the lowest unless otherwise specified in the product standard for the purpose of short term exposure assessment				